Huawei Access Node HEB-ACC-75 Routing Protocol Flap Resolution SOP

Node: HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5435

Description

The Huawei Access Node HEB-ACC-75, located at Hebbal Edge, is experiencing a Routing Protocol Flap due to BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. On the Huawei Access Node, BGP route churn is typically caused by improper configuration of the BGP neighbor or incorrect BGP network statement, leading to frequent advertisements and withdrawals of routes. OSPF adjacency loss can be attributed to incorrect OSPF network statement, mismatched hello intervals, or incorrect OSPF priority. ISIS LSP flooding is often a result of incorrect ISIS network statement, mismatched hello intervals, or incorrect ISIS priority. In this scenario, it is essential to verify the BGP neighbor configuration, OSPF network statement, and ISIS network statement on the Huawei Access Node to resolve the Routing Protocol Flap. Additionally, ensure that there are no hardware issues, such as faulty interfaces or power supply problems, that could be contributing to the failure domain.

Pre-requisite Checks

- Verify that the power supply and interfaces are functioning correctly on the Huawei Access Node HEB-ACC-75.

- Check the redundant link status to ensure that they are not causing the Routing Protocol Flap.

- Dispatch remote hands to Hebbal Edge to verify the site's environmental conditions.

Potential Root Causes

Cause	Probability
BGP Neighbor Configuration Issue	High
OSPF Network Statement Mismatch	Medium
ISIS Network Statement Error	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp neighbor, Expected Output: BGP Neighbor Information, Description: This command displays information about the BGP neighbor, including the neighbor IP address, state, and timer information. It helps to verify the BGP neighbor configuration on the Huawei Access Node.

Command: display ospf neighbor, Expected Output: OSPF Neighbor Information, Description: This command displays information about the OSPF neighbor, including the neighbor IP address, state, and timer information. It helps to verify the OSPF network statement on the Huawei Access Node.
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```
Command: display isis neighbor, Expected Output: ISIS Neighbor Information, Description: This command displays information about the ISIS neighbor, including the neighbor IP address, state, and timer information. It helps to verify the ISIS network statement on the Huawei Access Node.
```

Resolution Steps

Step 1: Step: Verify BGP Neighbor Configuration, Command: display bgp neighbor, Description: Verify the BGP neighbor configuration on the Huawei Access Node. Check for any errors or issues with the BGP neighbor.

Step 2: Step: Verify OSPF Network Statement, Command: display ospf network, Description: Verify the OSPF network statement on the Huawei Access Node. Check for any errors or issues with the OSPF network statement.

Step 3: Step: Verify ISIS Network Statement, Command: display isis network, Description: Verify the ISIS network statement on the Huawei Access Node. Check for any errors or issues with the ISIS network statement.

Post-Check Verification

```
Command: display bgp neighbor, Expected Output: BGP Neighbor Information, Description: Verify the BGP neighbor configuration on the Huawei Access Node.

Command: display ospf neighbor, Expected Output: OSPF Neighbor Information, Description: Verify the OSPF neighbor configuration on the Huawei Access Node.

Command: display isis neighbor, Expected Output: ISIS Neighbor Information, Description: Verify the ISIS neighbor configuration on the Huawei Access Node.
```

Roll-back Procedures

- Step: Restore Original BGP Neighbor Configuration, Command: undo bgp neighbor , Description: Restore the original BGP neighbor configuration on the Huawei Access Node.

- Step: Restore Original OSPF Network Statement, Command: undo ospf network , Description: Restore the original OSPF network statement on the Huawei Access Node.

- Step: Restore Original ISIS Network Statement, Command: undo isis network , Description: Restore the original ISIS network statement on the Huawei Access Node.